

Research Report for Mining, Deforestation, and Precipitation in the Brazilian Amazonas

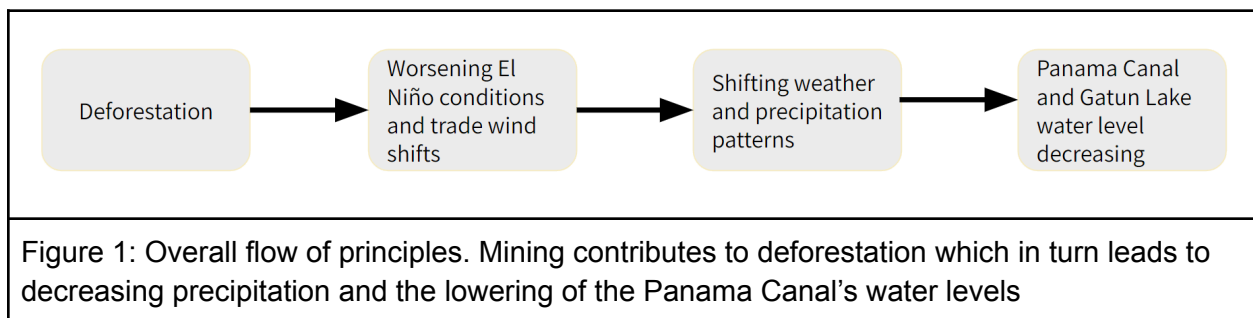
1. Introduction

The dual challenges of mining and deforestation have emerged as critical global concerns, drawing widespread attention from both consumers and companies. These interconnected issues have far-reaching and devastating consequences, including trade restrictions, infrastructure damage from increasingly severe weather events, and widespread crop failures caused by prolonged droughts and shifts in atmospheric convection patterns. Mining, a cornerstone of industrial development, comes with significant environmental costs. The extraction of minerals often leads to habitat destruction, soil erosion, water contamination, and the release of hazardous pollutants into the atmosphere. Additionally, mining activities frequently contribute to greenhouse gas emissions, further exacerbating climate change. The degradation of ecosystems and depletion of natural resources leave lasting scars on the environment, disrupting local biodiversity and jeopardizing the livelihoods of communities dependent on these landscapes. Similarly, deforestation accelerates these impacts by stripping the planet of its natural carbon sinks, further amplifying climate instability. Forest loss also disrupts weather patterns, reduces soil fertility, and undermines water cycles, compounding the damage initiated by mining practices. The cascading effects are evident across economic and social systems worldwide. Trade limitations arise as nations implement stricter environmental regulations to curb unsustainable practices, impacting global supply chains. Extreme weather events, such as floods and droughts, devastate communities, while agricultural systems face increasing challenges from soil degradation and water scarcity.

The Panama canal is one of the world's most used canals and an essential mainspring in global trade and maritime transportation. The Panama Canal offers a much safer and preferred shortcut from the Drake Passage, infamously dangerous for the rough mix of warm and cold water from three oceans, and currents, in addition to wind at that latitude meeting no resistance from any landmass, anywhere on the planet. The canal's main water source comes from the neighboring Lake Gatún, an artificial reservoir that feeds into Panama Canal's locks. As of September of 2023, under normal conditions, up to 38 ships pass through the canal per day(1), meaning that per year, around 13870 ships pass through the canal, contributing to a revenue of nearly \$4.97 billion per year (2). As the Panama Canal offers a crucial route for transportation connecting the western and eastern hemispheres of the globe, we must consider factors that limit its capability. Since early 2023, Panama and other local regions are currently being affected by an extended drought, leading to reduced rainfall, such as the 43% lower than average rainfall in October of 2023 (3), the results of which led to a reduced number of ships passing through the Panama Canal, such as from 32 ships to 31 ships on average passing through the canal in August of 2023 (1). The primary reason for the droughts affecting the Panama Canal is largely believe to be due to worldwide weather phenomenon called El Niño, a name for a weather phenomena that causes the Pacific jet stream (narrow, fast moving air flowing from East to West across the Central and North American regions) to move south and stretch further east. In recent years, the state of the trade wind shifts have grown more extreme, prompting it's outcomes, such as droughts, to similarly follow suit.

Convection currents are defined as heat-driven cycles that occur in the air, ocean, and mantle. These cycles are driven due to differing proximities to a heat source. The closer it is to the heat source the less dense the fluid, in this case air, gets as the molecules get more energized and the intermolecular forces weaken, increasing the volume of the fluid. The cycle then repeats with the less dense air rising above into the atmosphere, then cooling down, becoming more dense, and sinking back down to the heat source again due to pressures and forces when the fluids of different densities are affected by gravity pulling them closer to the center and thus the surface of the Earth. In atmospheric convection, an increase in temperature of the heat source produces higher temperature air which causes a greater reduction in density, which in turn causes lower pressure, more fluid motion and a stronger convection cycle.

Mining, often illegally in the Amazon region, requires large areas of land, which often requires the deforestation of trees. Deforestation and deliberately setting trees on fire once the felled trees are dry enough to burn, causes the rise in surface temperature over time, not only due to direct factors affecting the land deforested such the removal of shade, but also due to the indirect release of CO₂ when burning trees, causing a rise in the amount of CO₂ in the atmosphere, amplifying the greenhouse effect, and raising the average surface temperature of the globe overall. The local rise in surface temperature for the regions of deforestation causes the surface temperature to rise significantly due to the lack of shade blocking thermal energy from the sun. The increasing surface temperature, which is one of the heat sources heating up the air, drives stronger convection of the air directly above the deforested areas. This stronger drawing changes the airflow pattern in the surrounding regions, which has an impact on surface temperatures, and precipitation patterns and other weather phenomena leading to droughts in some surrounding areas. For example, it possibly contributes to trade wind shifts as seen in worsening El Niño conditions. Figure 1 shows the overall flow of the principles using a flow chart.



In 2023, the total mining area in the Brazilian Amazonas region increased by around 9.99% from 1.67×10^8 square meters in 2022 to 1.96×10^8 square meters in 2023 (4). In this report, using a convolutional neural network, we will be demonstrating the contributions of mining area data to deforestation in the Brazilian Amazonas territory which we will correlate with regional precipitation patterns, surface temperature, and the impact it has on drought causing events such as the worsening of El Niño(5). Finally, we will discuss the possible economic impacts that the drought phenomenon have on global economy

2. Method

2.1. Tools used in this study (loss function, adam optimizer functions)

In this study, we used the Earthrise Enterprise mining detector model(4). The Mining detector model, as shown in Figure 2, is a lightweight convolutional neural network, which was trained to differentiate mines from other topographical features by feeding it hand-labeled examples of mines and other key features as they appear in Sentinel-2 satellite imagery. The model uses a 32x3x3 filter paired with a max pooling size of 2x2(4). The Adam optimizer (Adaptive Moment Estimation) was utilized as the primary optimizer of the model. The Adam optimizer fuses the RMSprop and Stochastic Gradient Descent with momentum algorithms(5).

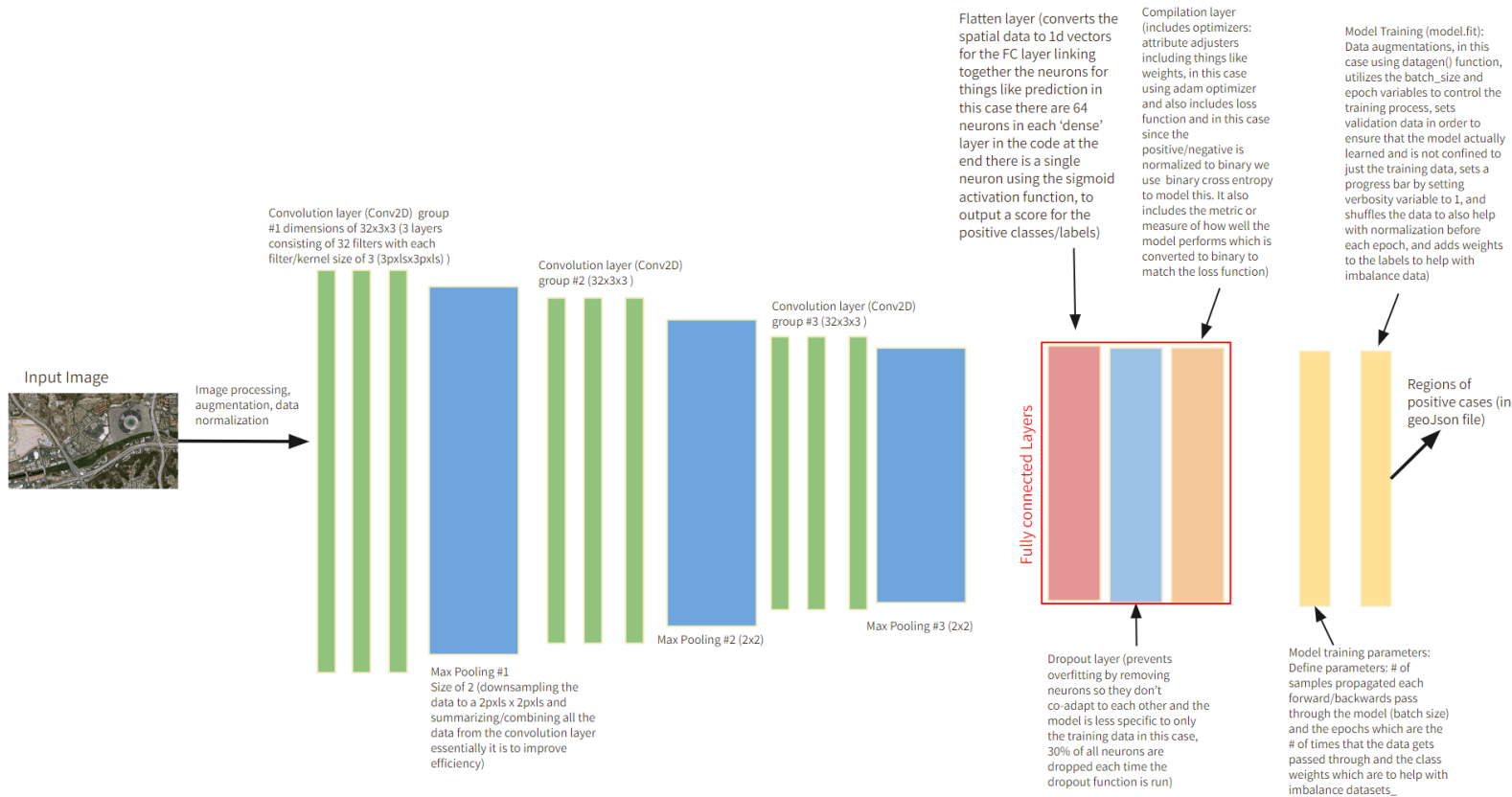


Figure 2: Structure of the CNN model (6)

In addition to the Earth Enterprise Mining Detector model, we also utilized the Google Earth Engine's Code Editor platform to gather numerical data on deforestation and precipitation(7).

2.2. Data Overview

The numerical geophysical and climactic data for the Brazilian Amazonas region include: 1) surface temperature, 2) total deforestation per year, 3) mining area per year, and 4) precipitation per year. The Sentinel-2 satellite imagery was used to train and run the convolution model for

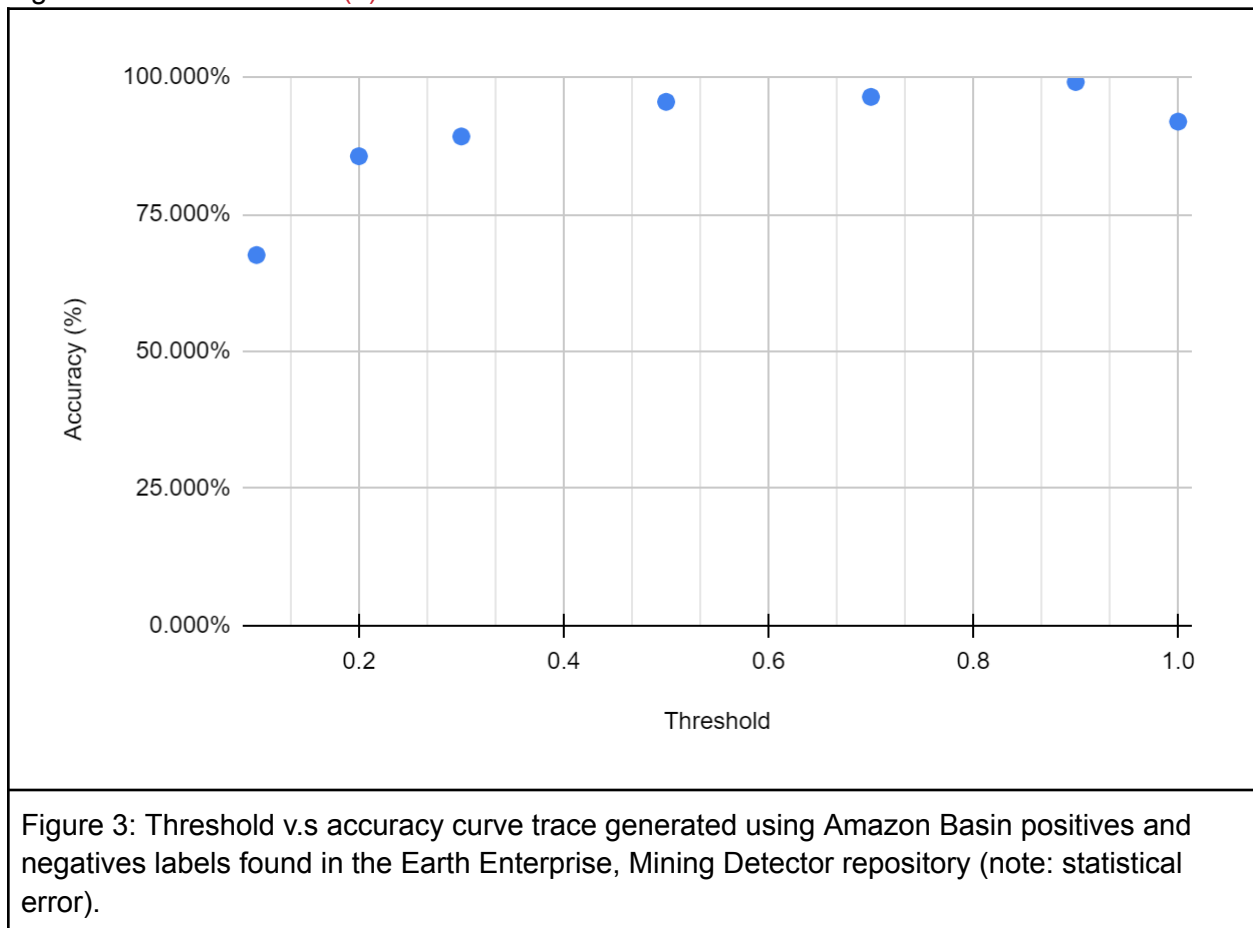
mining detection. The data is scaled at a 50m resolution, 13 spectral bands, 290 km swath, and a 5-day revisit time. Sentinel-2 data integrates two large visible near-infrared and shortwave infrared focal planes, combining 450 000 pixels(4). The Hansen Global Land Discovery and Analysis, and ECMWF Reanalysis v5 (ERA5) imagery data were used for the calculations of deforestation, average precipitation, and data respectively for the Brazilian Amazonas region(8,9). The Hansen Global Land Discovery and Analysis deforestation data is scaled at a resolution of 30.9 meters and utilized the treecover2000 band(8). The ERA5 precipitation data is scaled at a resolution of 11132 meters and utilized the total precipitation band(9).

3. Results (add detected mining areas on images; formulation of solution)

3.1. Earthrise Enterprise mining detector Model Performance Evaluation

We firstly study the accuracy of the model at different thresholds. We tested the model's accuracy against the labels (6) at thresholds of 0.1, 0.2, 0.3, 0.5, 0.7, 0.9, and 1.0.

Using the available labels for the Amazon Basin provided in the repository trimmed to fit within the boundaries of the Brazilian Amazonas, the Earthrise Enterprise mining detector model has an optimal threshold cut-off at 0.9. Figure 3 shows the threshold vs. accuracy chart generated for the model(6).



We generated a confusion matrix for the model shown in Figure 4. We rean the model at a threshold of 0.5 and compared the outputs against teh labels found in the Earth-Enterprise Repository. The model utilizes a binary classification of true and false labels and predictions with 1 representing true and 0 representing false.

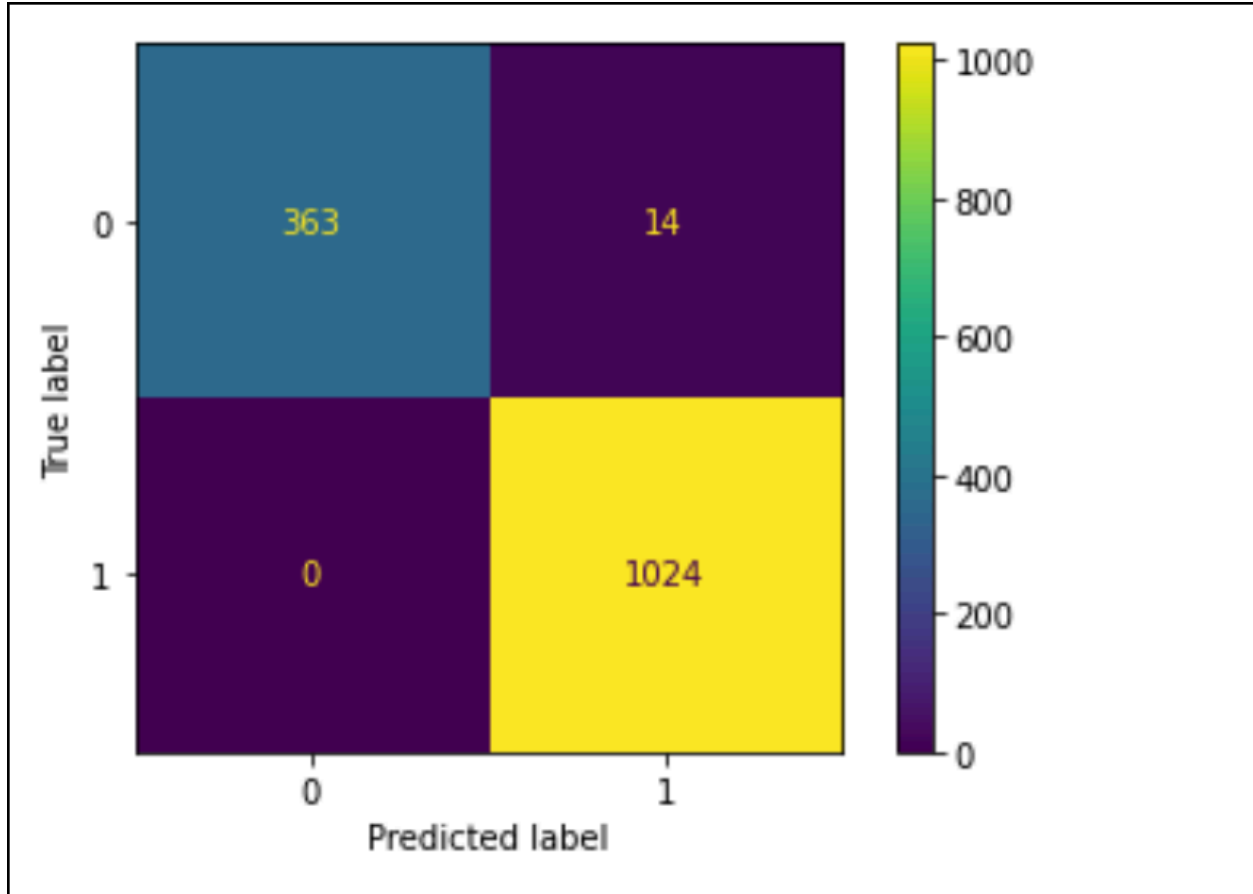
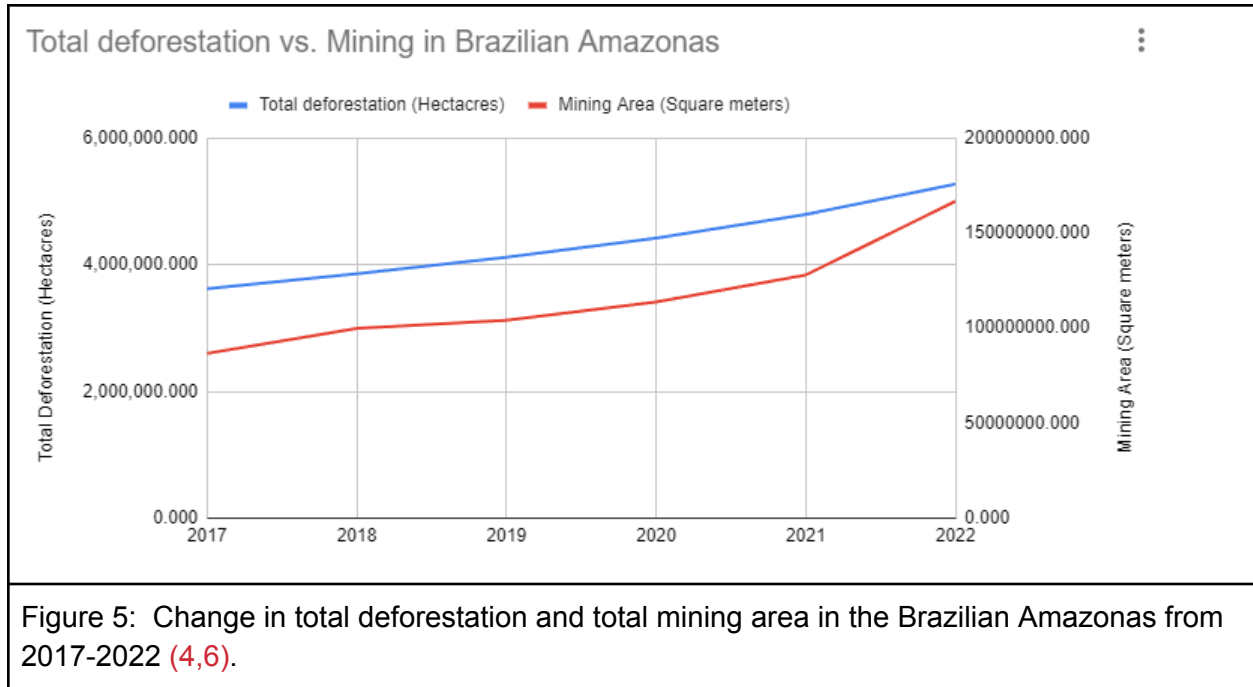


Figure 4: Confusion Matrix for the Earth Enterprise Mining Detector model. On the axis: the zeros represent negative predictions (x-axis) and labels (y-axis) while the ones represent positive predictions and labels (ie. (1,1) would be a true positive but (1,0) would be a false positive (0,0) would be a true negative while (0,1) would be a false negative). The color spectrum represents the amount of each label starting from 0-200 (purple) to 400-600 (blue) then finally to >1000 (yellow).

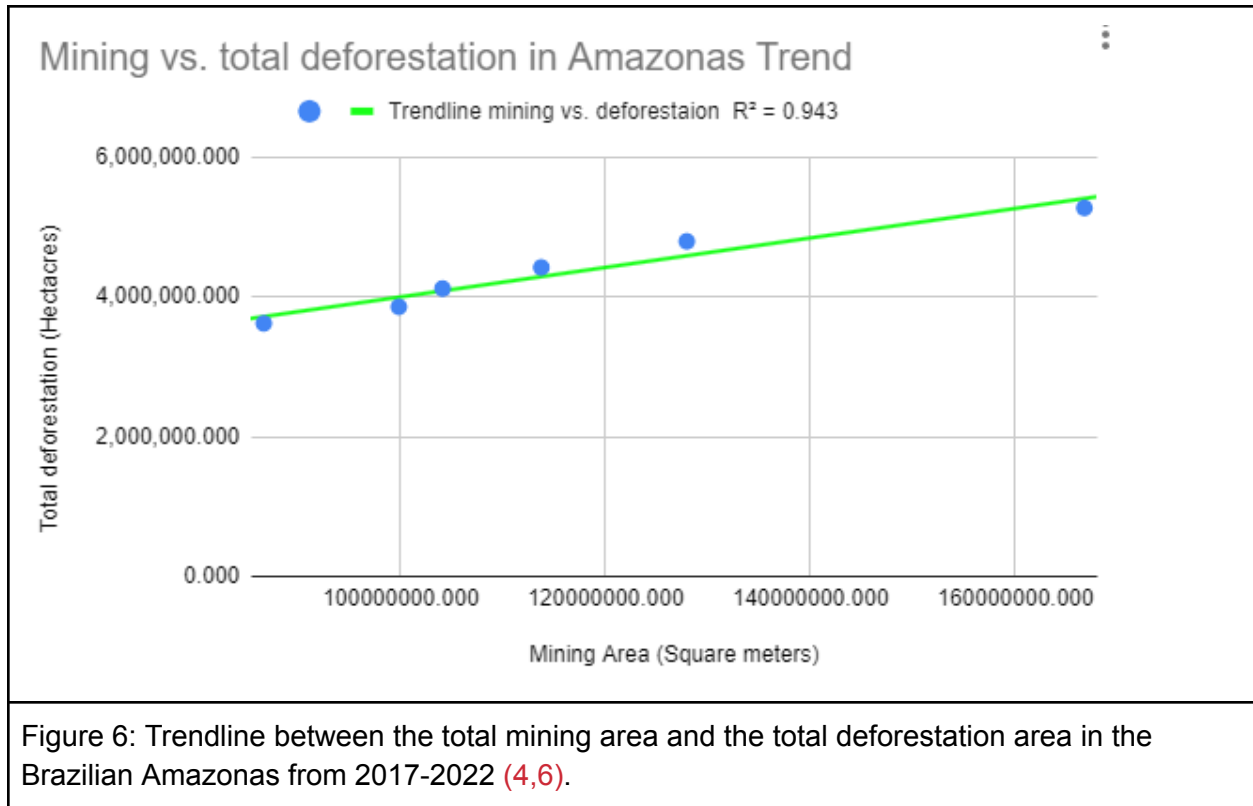
3.2. Data analysis

We independently plotted the trend of deforestation and total mining area in the Brazilian Amazonas from 2017 to 2022 in Figure 5. The deforestation data is from extracted from Hansen Global Forest database(8) and the total mining area is predicted by Earth Enterprise Mining Detector model using a threshold of 0.5. Overall, according to Figure 5, from 2017-2022 both the mining and the total deforestation within the Brazilian Amazonas region increased by 8.0×10^7 square meters and 1.7×10^6 hectares, respectively (4,6). Brazil holds around 60% of the

entire Amazon Rainforest, a lot of the mining within the Brazilian Amazonas (within the Amazon Rainforest region) contributes to the rising areas of deforestation. There is a positive correlation between mining and deforestation, showing the rising area of forest cleared contributed by the building of mines from 2017-2022.



In figure 6, we plotted a trendline between the Mining area versus the Total deforestation area. The deforestation data is from extracted hansen global data and Earthrise Mining Enterprise Repository (8) and the total mining area is predicted by Earth Enterprise Mining Detector model using a threshold of 0.5.



In figure 7, we independently plot the trend of deforestation over time and the trend of precipitation over time. For the total deforestation data we utilized the Hansen Global Forest Change database, and we obtained the precipitation data from the ERA5 Monthly database. In Figure 7, the precipitation from 2017 to 2022 has overall decreased by around 112589512 millimeters per month (11). Overall there is a negative correlation between the increase of deforestation and the amount of precipitation. Our results show that there is a correlation between the amount of deforestation in the Brazilian Amazonas and the decrease in precipitation in the local region around it (13). (10) Since precipitation and other cloud-related weather phenomenon are largely driven by wind and the movement of air, the increase in deforestation, contributed by mining, must have altered the movement of the air driving many precipitation related phenomenon like the droughts driven by El Niño affecting Lake Gatún Panama Canal (3) with overall lowered water levels from 2017-2023, contributing to deteriorating status of global economic effects.

Figures 5 and 6 show the change in total area of mining compared to that of deforestation in the Brazilian Amazonas, and the trendline between them respectively. Figure 6 and 7 show the change in the total area of mining in the Brazilian Amazonas compared to the total precipitation amount in millimeters, and the trendline between the two, respectively. The data for the Gatun lake water levels was collected by observation from the Gatun H₂O Official Gatun Water Levels(12).

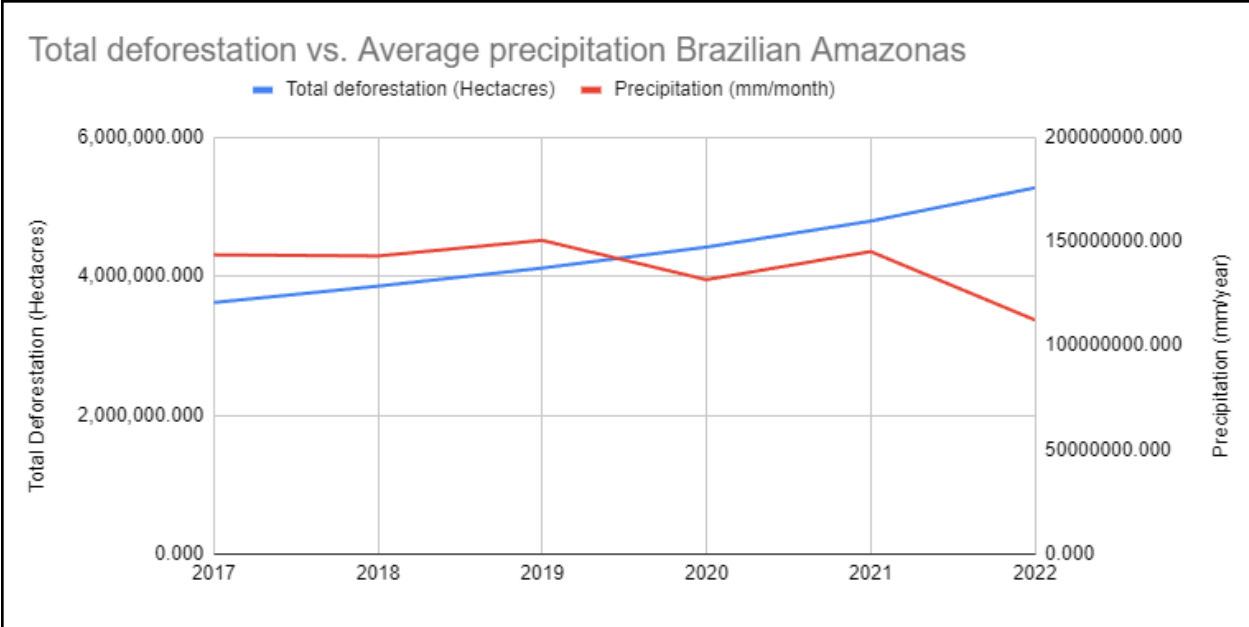


Figure 7: Change in total mining area and change in average precipitation in millimeters in the Brazilian Amazonas region from 2017-2022 (4,11).

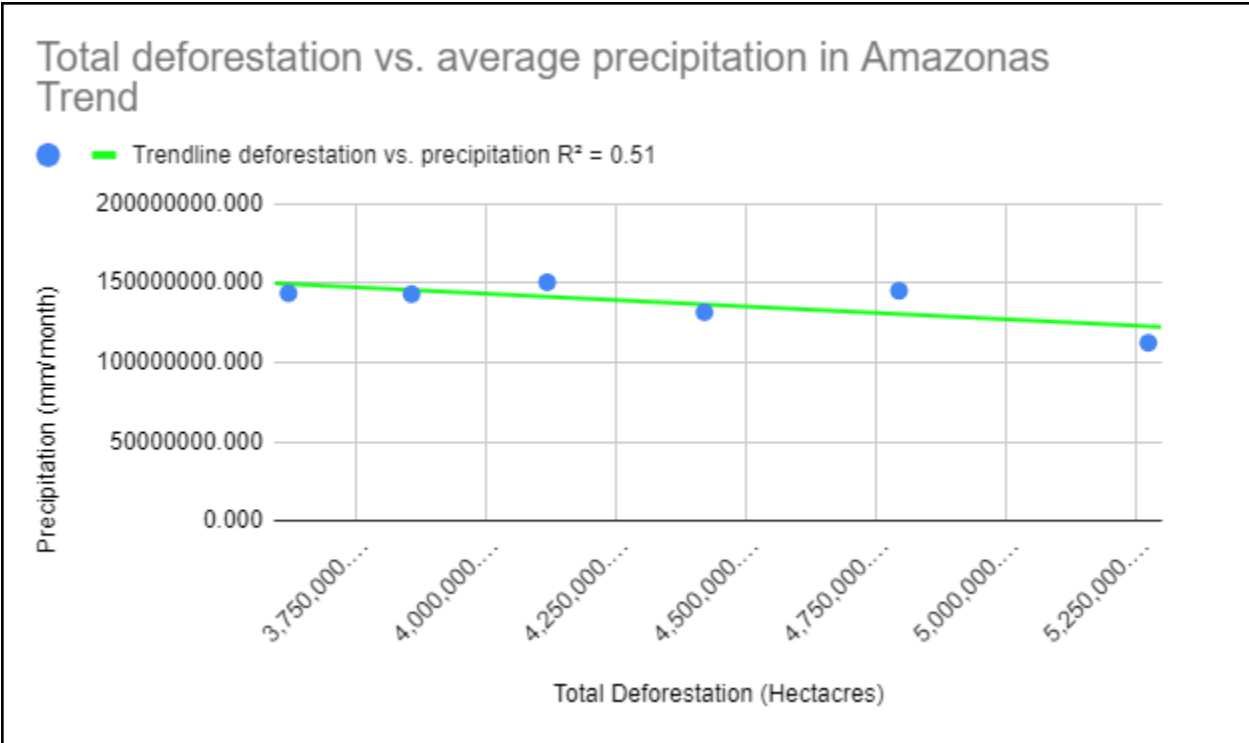


Figure 8: Trendline between total deforestation area and average precipitation amount in millimeters in the Brazilian Amazonas region from 2017-2022 (4,11).

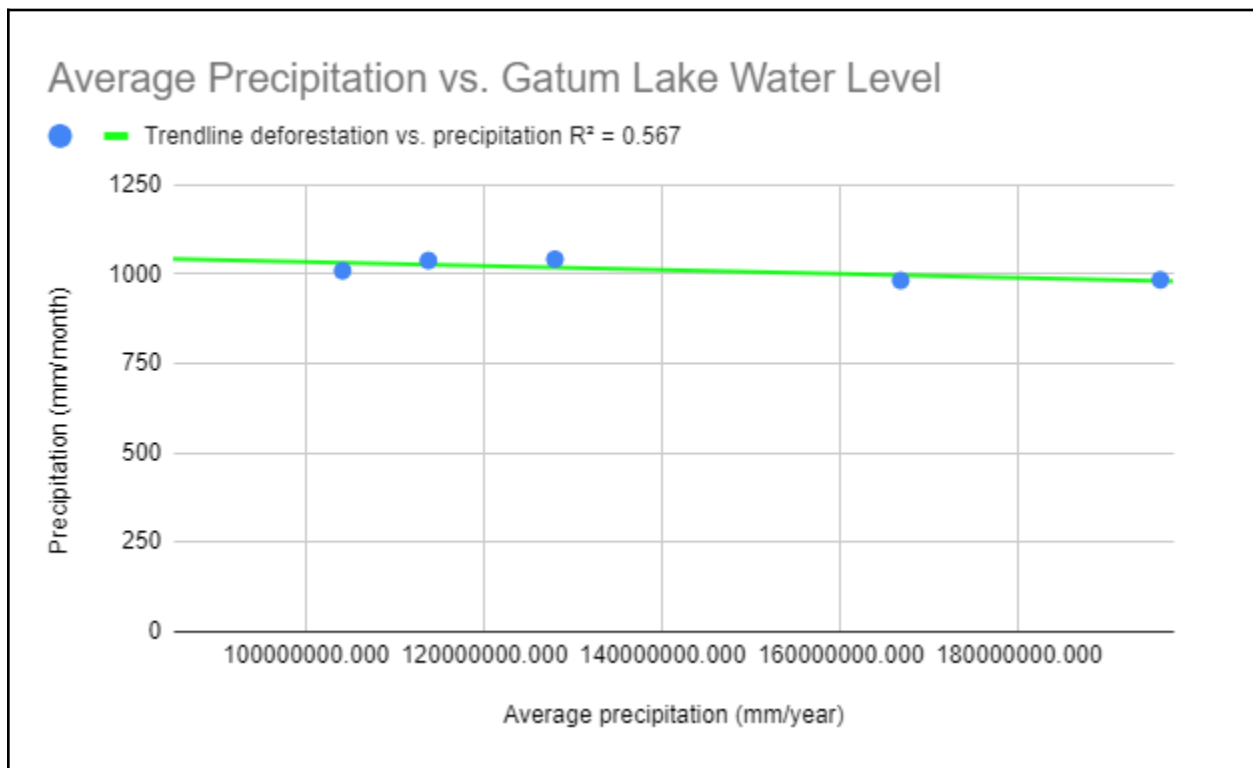
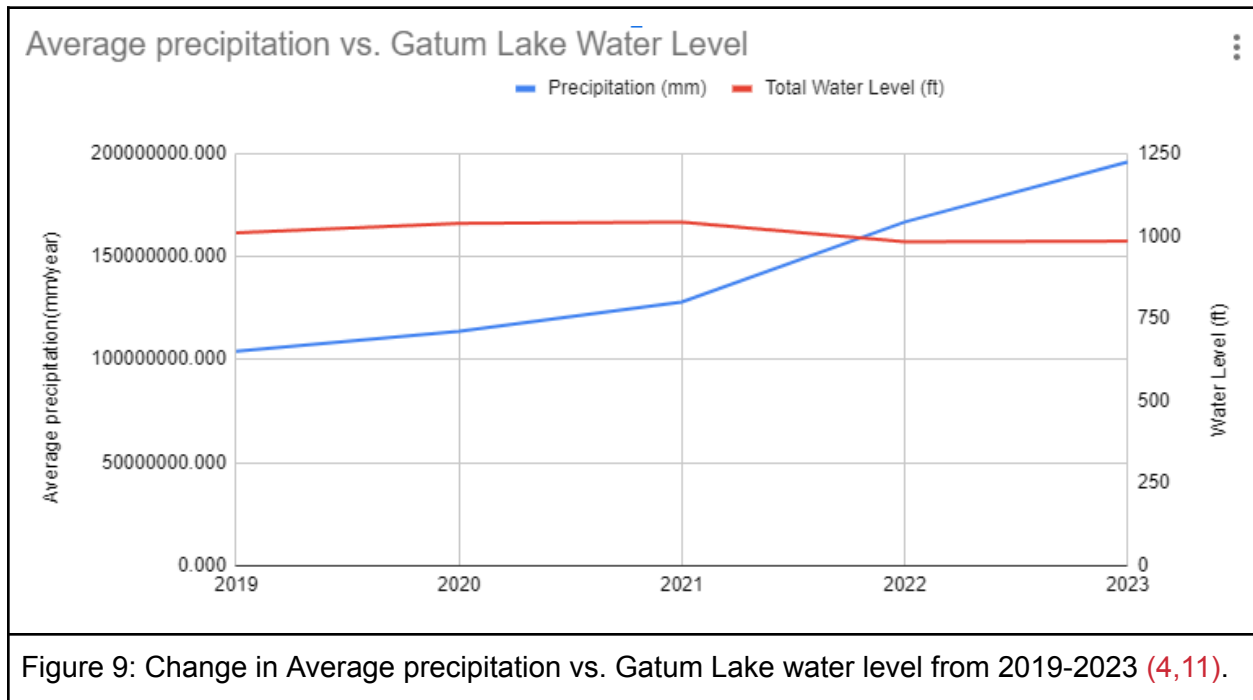


Figure 10: Correlation between Average precipitation and Gatun Lake water levels from 2017 to 2023 (4,11).

4. Conclusion

The results from the study have indicated that there is a positive correlation between mining and deforestation, a negative correlation between deforestation and precipitation amounts in local areas, and a correlation between the decreased water levels at Gatun Lake and the increase and decrease of precipitation. (include methodology and other discoveries; hyperparameters, limitations of study)

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